Predicting position and uncertainty

LEARNING TARGET · R3 I can use velocity and time to predict where it will be, including the possible error.

SECTION A KEY VOCABULARY

Starting position	Where the object is at t = 0. It is often not zero.
Change	Velocity × time: how far the position shifts.
Final position	Start + change.
Prediction range	A ± bound. "10 ± 2 m" means 8 m to 12 m. More time usually means a wider range.

SECTION B KEY EQUATIONS + UNITS

$$x_f = x_0 + v\,t$$

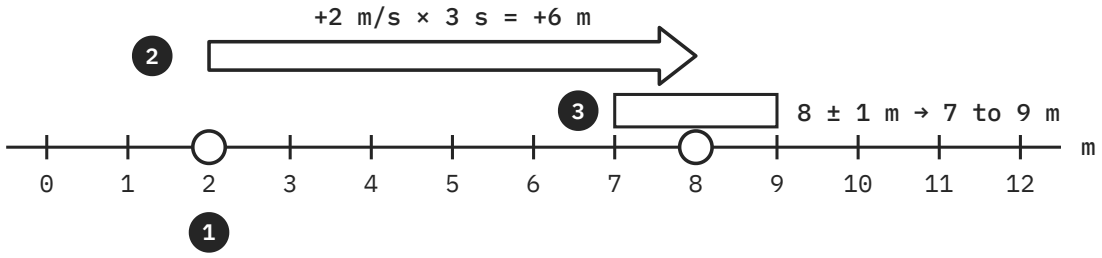
largest error = start error + (velocity error × time)

units first 2 h = 7,200 s

2-D also $y_f = y_0 + v_y t$

SECTION C COLOR IT · START, CHANGE, RANGE

Start 2 m, +2 m/s for 3 s.



☐ **1 · BLUE** Starting position (dot)

☐ **2 · BLUE** The change, +6 m (arrow)

☐ **3 · BLUE** The range, 7 to 9 m (light shade)

SECTION D WORKED EXAMPLE

A cart starts at 8 m and moves at −2 m/s for 3 s. Where is it at the end?

GIVEN	EQUATION	WORK	ANSWER
x_i = 8 m v = −2 m/s t = 3 s	$x_f = x_i + v \times t$	change = (−2)(3) = −6 m x_f = 8 − 6	**2 m** moved 6 m left

SECTION E THE 2026-XJ THREAD

DAY 5 OF 22 · NEW TOOL: FORECAST + UNCERTAINTY

With a starting position and a velocity, we can forecast where XJ will be. But every measurement has error, so a responsible forecast is a range, not a single point.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else